INFRASTRUCTURE & NETWORK

Infrastructure & Network Security

AI-powered assessment of your network perimeter, internal infrastructure, and Active Directory environments.

Inside This Document

Key Benefits • Your Challenge • Services Overview • The Technieum Edge • Deliverables

KEY BENEFITS

Why This Matters

360° Attack Surface Visibility

AI-Based ASM maps every external asset using 100+ orchestrated reconnaissance tools across four phases.

Attack Path Intelligence

Individual vulnerabilities chained into multi-step compromise paths demonstrating real business impact.

Zero False Positives

AI accelerates discovery; human experts validate every finding. You act on real, exploitable risk only.

YOUR CHALLENGE

Your enterprise infrastructure spans external perimeters, internal networks, Active Directory, wireless access points, and cloud boundaries. This creates:

- Expanding attack surfaces with shadow IT and forgotten assets
- Complex AD environments with hidden privilege escalation paths
- Segmentation gaps between security zones
- Configuration drift across servers and network devices
- Need for a partner who validates real exploitability, not scanner noise

Why Technieum as Your Trusted Partner

Technieum brings 6 proprietary AI platforms, an in-house R&D team actively hunting zero-days, and AI-powered methodologies that deliver validated, exploitable findings with guaranteed zero false positives — enabling your team to focus remediation effort where it matters most.

OUR SOLUTION

Services Overview

Technieum deploys AI-powered reconnaissance, cross-correlated testing pipelines, and human-validated exploitation across every infrastructure layer.

1	**Attack Surface Management**	Continuous discovery and monitoring of all external-facing digital assets using our AI-Based ASM platform. 100+ orchestrated reconnaissance tools map subdomains, IP ranges, exposed APIs, cloud resources, shadow IT, and forgotten infrastructure across four phases: discovery, DNS resolution, deep analysis, and vulnerability validation.
2	**Internal Network Testing**	Simulates an insider threat or compromised endpoint scenario within your internal environment. Tests lateral movement paths across VLANs, service exploitation, credential harvesting via Responder/MITM techniques, privilege escalation to domain admin, and identifies how far an attacker can move once inside your perimeter.
3	**External Network Testing**	Evaluates internet-facing infrastructure — firewalls, VPNs, web servers — for exploitable vulnerabilities.
4	**Active Directory Audit**	Comprehensive AD security assessment using Technieum's AD Suite — Kerberoasting, AS-REP Roasting, DCSync, Golden/Silver Ticket attacks, GPO abuse, ACL exploitation, delegation attacks, certificate service abuse (ESC1-ESC8), and full privilege escalation chain mapping from standard user to domain admin.
5	**Wireless Security Testing**	Rogue AP detection, WPA2/WPA3 testing, evil twin attacks, PMKID capture, guest network isolation.
6	**Build & Config Review**	Systematic review against CIS Benchmarks — OS hardening, service configs, patch levels, policy enforcement.
7	**Segmentation Testing**	Validates network segmentation between PCI, OT/IT, guest, and critical asset zones.
8	**Attack Path Mapping**	Chains vulnerabilities into multi-step attack paths from perimeter to critical asset compromise.

THE TECHNIEUM EDGE

Why Technieum for Infrastructure

Our Methodology
Every engagement runs five converging pipelines — ASM, SAST, AD Suite, ToIP, and Nuclei — all cross-correlated via the OffSec Management Portal.

AI-Based ASM Platform
100+ recon tools across 4 phases — capabilities no scanner provides.

AD Suite
Purpose-built AD attack platform — Kerberoasting, DCSync, Golden Ticket, domain compromise.

ToIP Intelligence Engine
15,000+ real-world exploits via RAG search — instant playbooks with exact payloads.

360° Cross-Correlation
ASM + SAST + AD + network findings unified for complete visibility.

Zero False Positives
AI finds, humans validate every finding before reporting.

OffSec Management Portal
Real-time dashboards, multi-tester collaboration, one-click reporting.

Standards Alignment

Every Technieum engagement aligns with globally recognized security frameworks including OWASP Top 10, OWASP API Top 10, OWASP LLM Top 10, PTES, NIST CSF, NIST AI RMF, MITRE ATT&CK, CIS Benchmarks, IEC 62443, and ISO 27001:2022. Our team holds OSCP, OSEP, CRTE, CRTP, eWPTX, eCPPT, and CEH certifications. Technieum is ISO 9001:2015 and ISO 27001:2022 certified.

OUR METHODOLOGY

How We Deliver Results

Technieum's infrastructure security methodology follows a structured five-phase approach tailored to network and infrastructure environments — ensuring comprehensive coverage from perimeter to Active Directory core.

Discover	Analyze	Attack	Report	Remediate
Asset enumeration	ToIP intelligence	Manual exploitation	CVSS/CWE scoring	AI-powered fixes
Tech fingerprinting	Logic extraction	Business logic test	PoC evidence	Stack-specific code
OSINT gathering	Threat modelling	Chain validation	Exec summary	Config guidance
Scope validation	Risk prioritization	Privilege escalation	Test case docs	Retest validation
Port scanning	Attack planning	Lateral movement	Playbook docs	Closure sign-off
Service detection	Playbook generation	Data exfiltration	Risk heatmap	Knowledge transfer

Phase 1 — Discover

AI-Based ASM deploys 100+ reconnaissance tools for external asset discovery. Internal network enumeration maps VLAN structures, IP ranges, and service endpoints. AD enumeration identifies domain controllers, trust relationships, and GPO configurations.

Phase 2 — Analyze

ToIP intelligence engine generates infrastructure-specific attack playbooks. AD Suite maps privilege escalation paths and domain compromise chains. Network topology analysis identifies segmentation weaknesses and lateral movement opportunities.

Phase 3 — Attack

Manual exploitation of network services, AD attack chains (Kerberoasting, DCSync, Golden Ticket), wireless security testing, VPN bypass attempts, and segmentation validation through controlled lateral movement.

Phase 4 — Report

Technical reports with network diagrams, AD attack path visualizations, CVSS-scored findings with PoC evidence, and infrastructure-specific risk heatmaps. Management summaries contextualize risk for leadership.

Phase 5 — Remediate

Infrastructure-specific remediation guidance — AD hardening configurations, network segmentation rules, firewall policy recommendations, and CIS Benchmark alignment steps. Followed by retest validation.

DELIVERABLES

What You Receive

Every engagement produces four comprehensive deliverables designed for different stakeholders — from technical teams requiring reproduction steps to executives needing business impact context.

Technical Report	Management Report	Threat Model Report	ToIP Playbook
CVSS-scored, CWE-mapped findings with proof-of-concept evidence and step-by-step reproduction instructions for every vulnerability discovered.	Executive-level risk summary with business impact analysis, prioritized recommendations, and strategic remediation roadmap for leadership.	Comprehensive threat landscape analysis with attack vectors mapped to MITRE ATT&CK framework specific to your environment.	Target-specific attack playbooks generated from 15,000+ real-world disclosed vulnerabilities via our RAG-powered intelligence engine.

- ✓ **Technical Report**
  Detailed findings with CVSS scoring, CWE mapping, proof-of-concept evidence, and step-by-step reproduction instructions.
- ✓ **Management Report**
  Executive-level risk summary with business impact analysis, prioritized recommendations, and strategic remediation roadmap.
- ✓ **Threat Model Report**
  Comprehensive threat landscape analysis specific to your environment, attack vectors, and adversary profiles mapped to MITRE ATT&CK.
- ✓ **ToIP Playbook Document**
  Target-specific attack playbooks generated from 15,000+ real-world disclosed vulnerabilities via our RAG-powered intelligence engine.

All deliverables are produced within the agreed timeline with milestone-driven tracking. Critical findings are escalated immediately via the OffSec Management Portal — you never wait for the final report to learn about urgent risks.

Ready to Secure Your Organization?

Partner with Technieum for AI-powered offensive security with zero false positives.

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